VLSI Devices
Lecture 1

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Course

- Course number: EC4313 & SE3101
- Credits: 3
- Schedule: 09:00~10:15 every Monday/Wednesday
- Instructor: Sung-Min Hong

Prerequiste and references

- Semiconductor Materials and Devices (EC3206) *or an equivalent course*
- Textbook
  - Y. Taur and T. H. Ning, Fundamentals of Modern VLSI Devices
  - **3rd** edition

Prof. Yuan Taur
(UCSD)

Resources

- Presentation materials
  https://github.com/hi2ska2/vlsidevices2026s

- Homework submission and notice
  - GIST LMS system

- YouTube channel
  https://www.youtube.com/@TCADHong

Grading and policy

- Attendance: 10 %
- Mid-term examination: 40 %
- Final examination: 50 % (**June 10**)

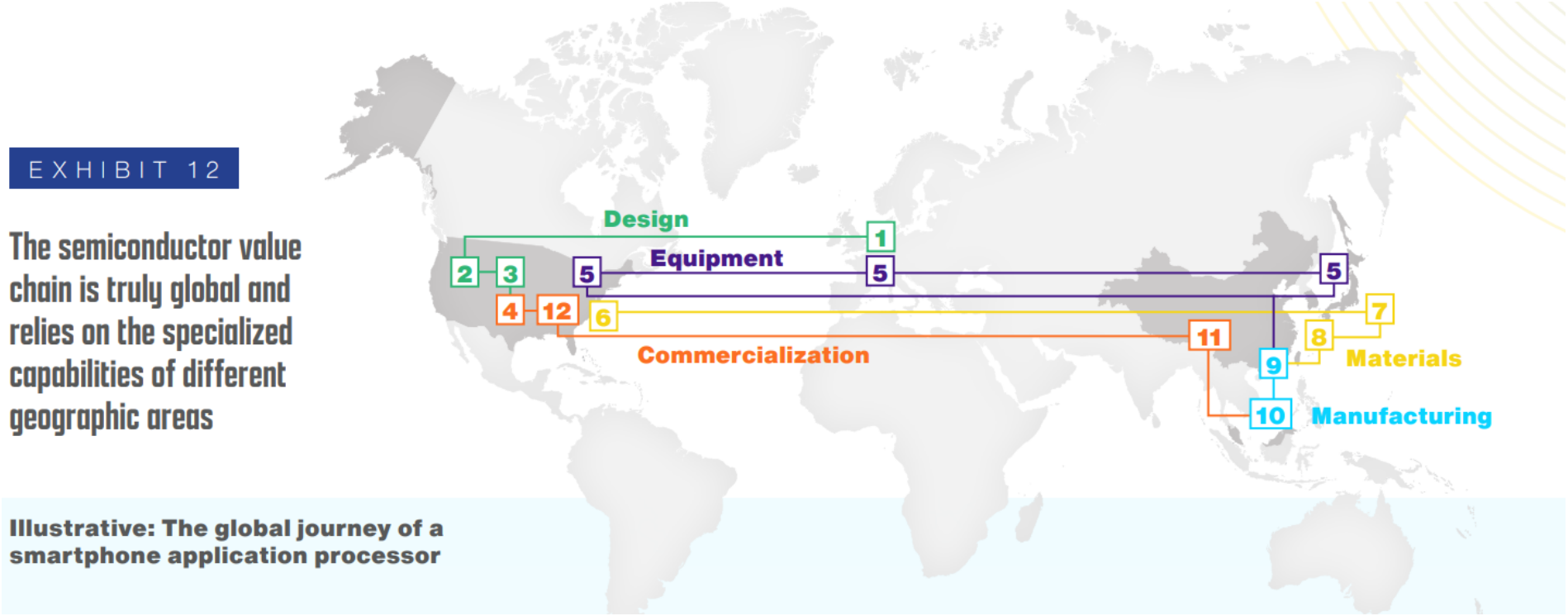
- Homework
  - Some assignements may be given. However, they will not contribute to the total score.

L1

Geopolitical dimensions

- The semiconductor industry is not just economically important.
  - Six major regions (US, South Korea, Japan, mainland China, Taiwan, and Europe)

(Semiconductor Industry Association)

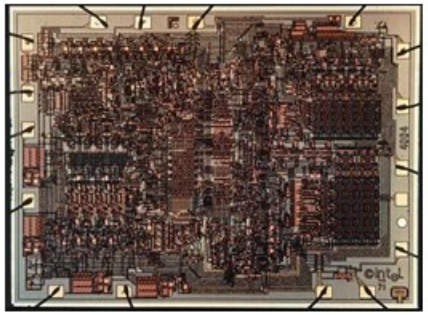


My own lecture material in 2014

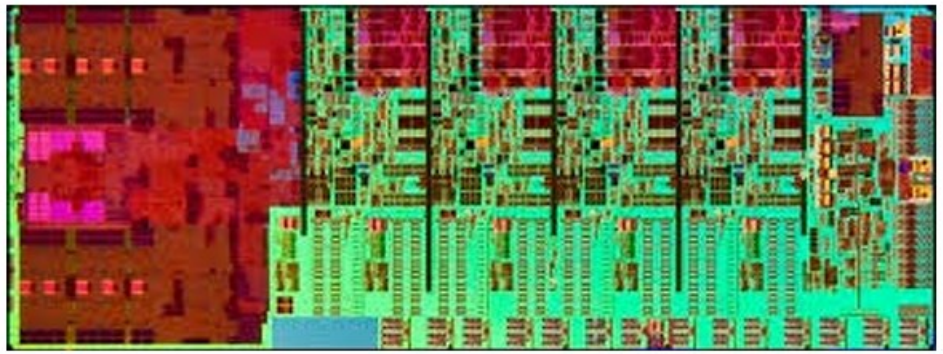
- “Digital design” course

Die shot

- More than 40 years between them



Die size: 12 sq mm
Min. feature size: 10 micron
Max. clock speed: 740 kHz



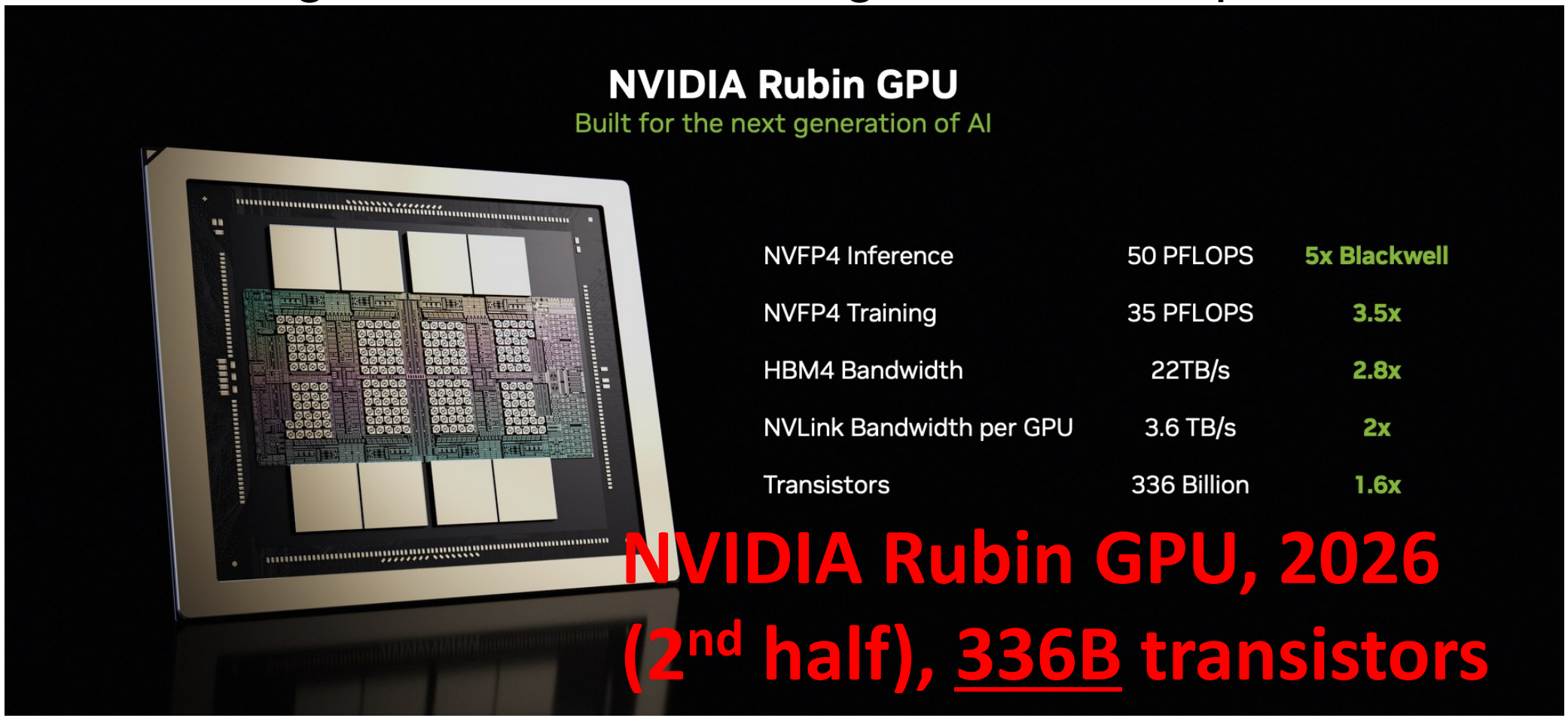
Die size: 177 sq mm
Min. feature size: 22 nm

Intel 4004, 1971, 2.3k transistors

Intel Core i7 4770K, 2013, 1.4B transistors

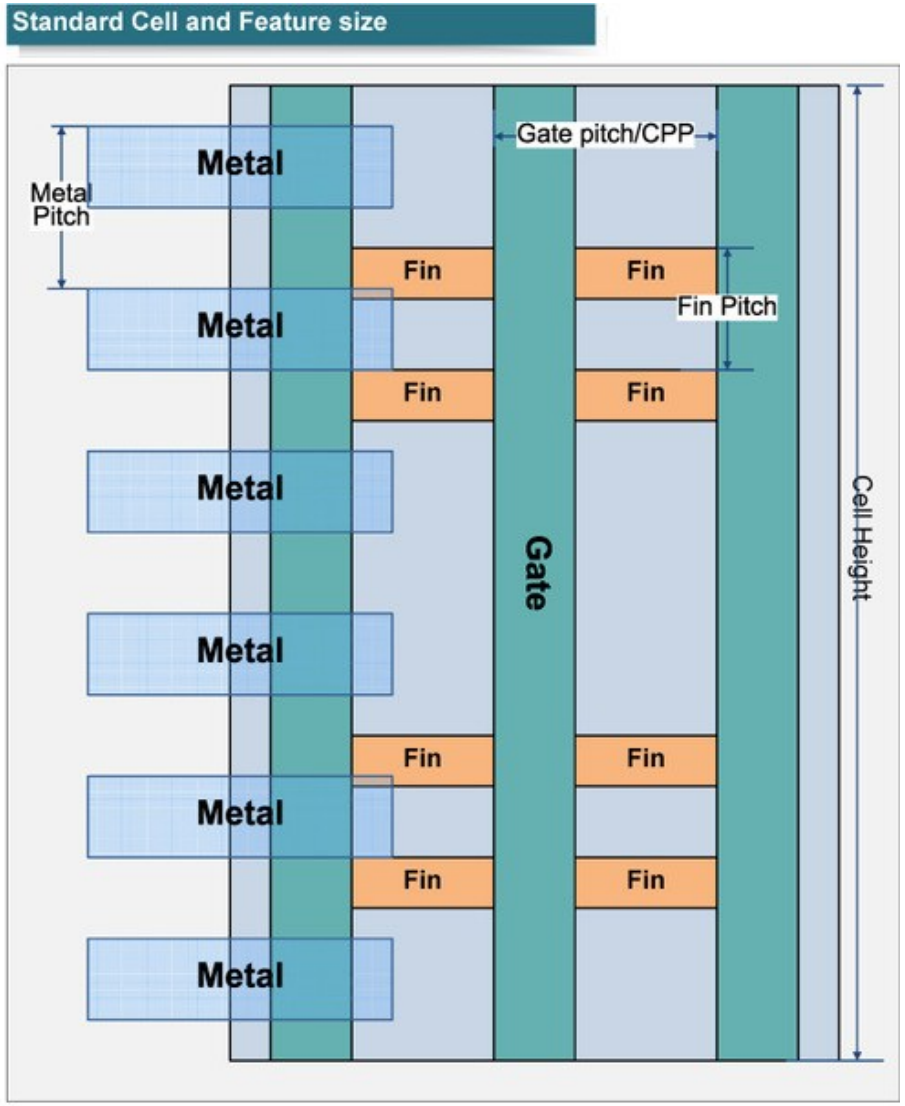
As of 2026,

- What is the state-of-the-art?
  - It is no longer a CPU. It is no longer an Intel chip.



What is the magic behind it?

- Increasing the die size helps a lot.
  - The main driver is the transistor scaling.
  - Intel 22nm: 16.5M transistors/mm^2 (Wikipedia, “22 nm process”)
  - TSMC N5: 138.2M transistors/mm^2 (Wikipedia, “5 nm process”)
  - TSMC N3E: 216M transistors/mm^2 (Wikipedia, “3 nm process”)



Standard cell and feature size
(H. Goto, PC Watch article, 2017)

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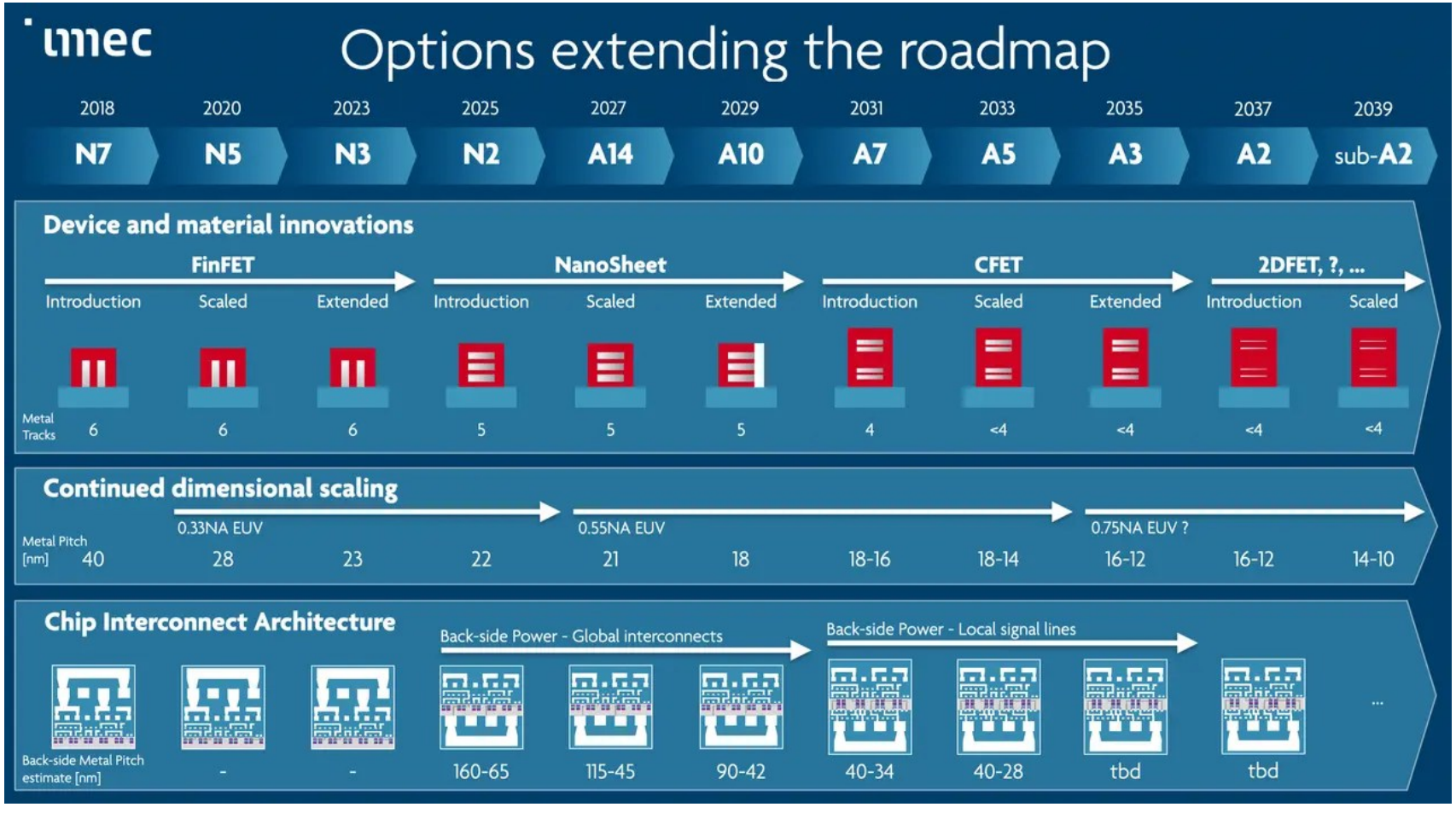
Scaling roadmap

- Transistor scaling is extremely difficult.
  - *"It ain't over till it's over."*



IMEC's roadmap

Poisson equation

- Maxwell's first equation (Gauss' law)
  - The electric displacement field, $\mathbf{D}$, satisfies
$$\nabla \cdot \mathbf{D} = \rho$$
  - The net charge density, $\rho(\mathbf{r})$, is $q[p(\mathbf{r}) - n(\mathbf{r}) + N_d^+(\mathbf{r}) - N_a^-(\mathbf{r})]$.
  - (q is the absolute electronic charge.)
  - In a linear, isotropic dielectric medium, is proportional to the electric field, $\mathbf{E}$,
$$\mathbf{D} = \epsilon \mathbf{E}$$
ϵ: permittivity
  - Therefore,
$$\nabla \cdot (\epsilon \mathbf{E}) = \rho$$

Electrostatic apppoximation

- The electric field can be expressed
  - In terms of a scalar potential, ϕ, and a vector potential, $\mathbf{A}$,

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}$$

  - In the electrostatic approximation,

$$\mathbf{E} = -\nabla\phi$$

Taur, ~Eq. (2.55)

  - Then,

$$\nabla \cdot (\epsilon \nabla \phi) = -\rho$$

  - When ϵ is spatially uniform,

$$\nabla^2 \phi = -\frac{\rho}{\epsilon}$$

Taur, ~Eq. (2.56)

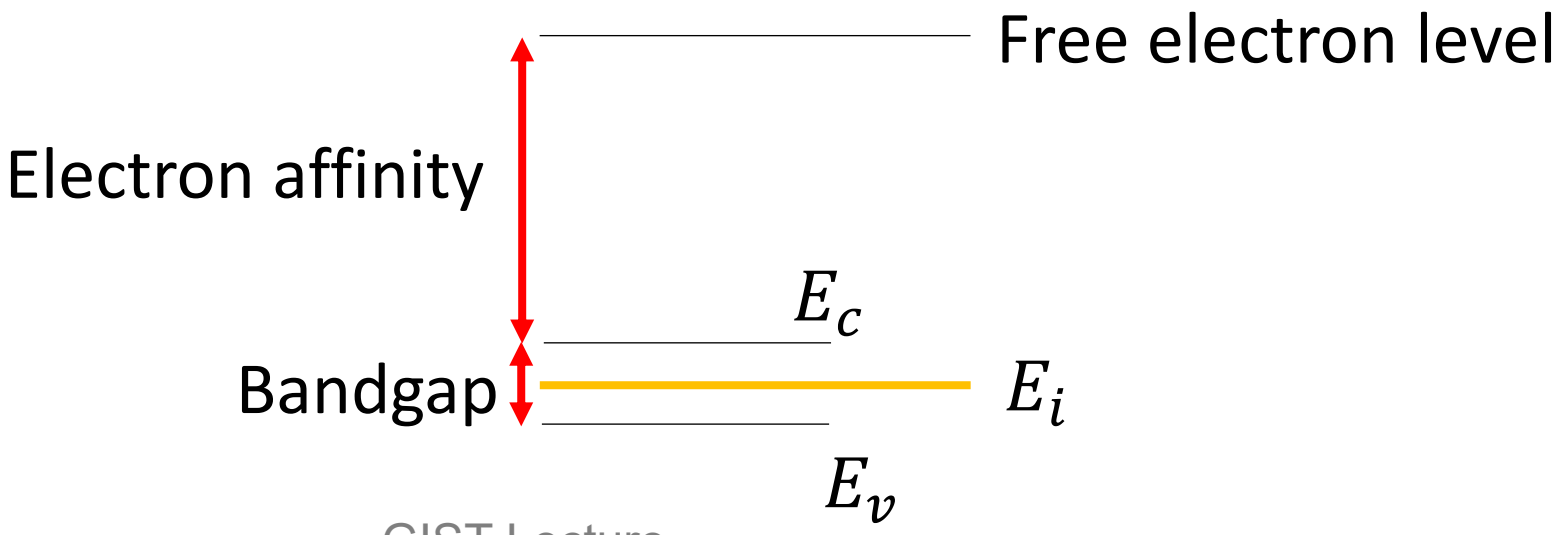
Convention for ϕ

- Conventionally, the electrostatic potential is defined in terms of the intrinsic Fermi level, E_i ,

$$\phi = -\frac{E_i}{q}$$

Taur, Eq. (2.54)

  – (q is the absolute electronic charge.)



Carrier densities at equilibrium

- Carrier densities are expressed as
  - Electron density

$$n = n_i \exp\left(-\frac{E_i - E_f}{k_B T}\right) = n_i \exp\left(\frac{\phi - \phi_f}{k_B T/q}\right)$$ Taur, Eq. (2.63)

  - The Fermi potential, ϕ_f, is $-\frac{E_f}{q}$.
  - Hole density

$$p = n_i \exp\left(\frac{E_i - E_f}{k_B T}\right) = n_i \exp\left(-\frac{\phi - \phi_f}{k_B T/q}\right)$$ Taur, Eq. (2.64)

  - These relations are generally applicable in the presence of net charge and band bending.

Thank you!